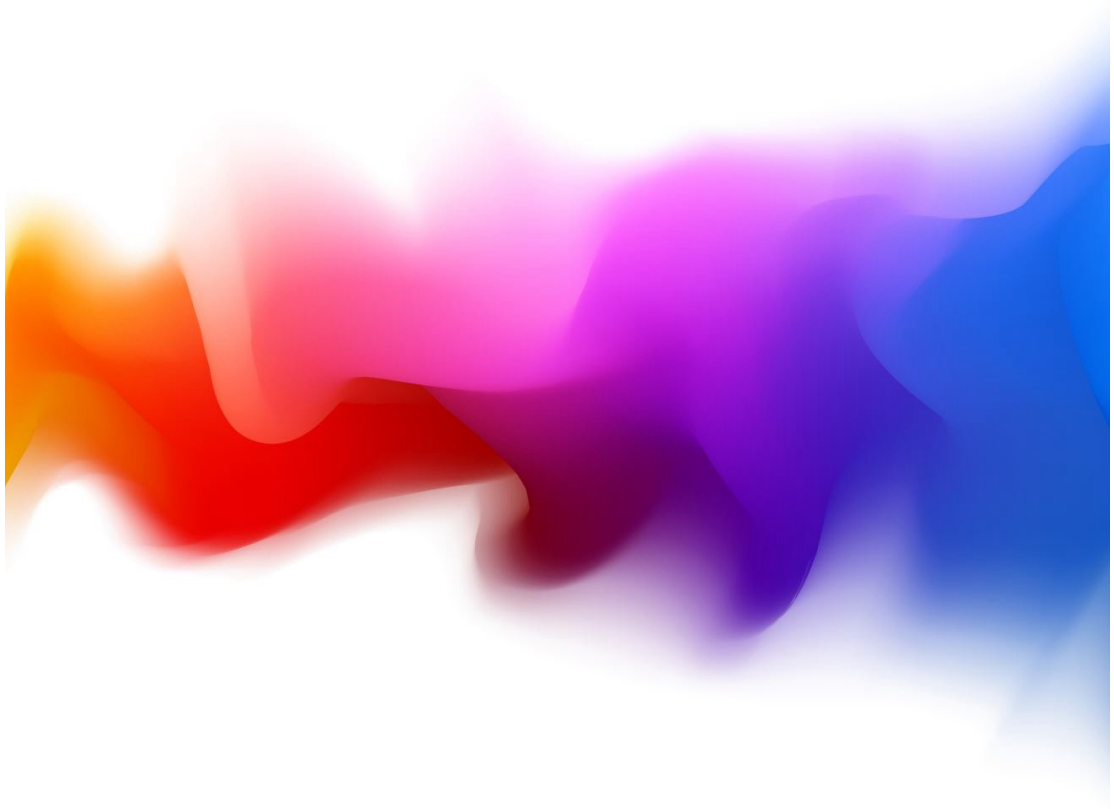


Medical AI in Global Context

COLLEGE 117

May 6, 2026

Dr. Martinez-Martin

Dr. Daphne Martschenko

Overview

- Bias & social context
  - Approaches for evaluating and mitigating bias
  - Defining principles

Approaches to mitigation

What does “bias” mean in AI/machine learning?

Bias in machine learning

"**Bias** is a **systematic error** that leads to an incorrect estimate of effect or association." (statsdirect)

"Bias is an **unfair** systematic error..." (Cho, 2021)

Does the AI work?

Faden & Powers- Key Questions

Would people without [this technology] be so deprived that they could not live decent lives as human beings?

Does not providing access lead to the kinds of disadvantages that are so systematically oppressive that people cannot have decent lives?

Obermeyer study: bias in healthcare algorithm

- "At a given risk score, Black patients were considerably sicker than White patients, as evidenced by signs of uncontrolled illnesses"
- Algorithm used cost as a proxy for level of sickness
- Black patients generated lower costs than white historically Black patients also underrepresented in the datasets
- If disparity fixed, percentage of Black patients receiving additional help would increase from 17.7 to 46.5%

AI & Dermatology



› Sci Adv. 2022 Aug 12;8(32):eabq6147. doi: 10.1126/sciadv.abq6147. Epub 2022 Aug 12.

Disparities in dermatology AI performance on a diverse, curated clinical image set

Roxana Daneshjou [1 2], Kailas Vodrahalli [3], Roberto A Novoa [1 4], Melissa Jenkins [1], Weixin Liang [5], Veronica Rotemberg [6], Justin Ko [1], Susan M Swetter [1], Elizabeth E Bailey [1], Olivier Gevaert [2], Pritam Mukherjee [2], Michelle Phung [1], Kiana Yekrang [1], Bradley Fong [1], Rachna Sahasrabudhe [1], Johan A C Allerup [1], Utako Okata-Karigane [7], James Zou [2 3 5 8], Albert S Chiou [1]

Affiliations + expand

PMID: 35960806 PMCID: PMC9374341 DOI: 10.1126/sciadv.abq6147

- Estimated 3 billion people worldwide have inadequate access to medical care for skin conditions
- AI tools could help with screening and diagnosis of skin conditions
- Databases most commonly used for dermatology AI
  - ModelDerm trained on mix of images from white and Asian patients
  - DeepDerm and HAM10000 trained primarily on white patient images
- These models had substantial limitations when applied to Diverse Dermatology Images (DDI) dataset

The Fitzpatrick Skin-Type Chart

Genetic Disposition					
Score	**0**	**1**	**2**	**3**	**4**
What is the colour of your eyes?	Light blue, Grey, Green	Blue, Grey or Green	Blue	Dark Brown	Brownish Black
What is the natural colour of your hair?	Sandy Red	Blond	Chestnut/ Dark Blond	Dark Brown	Black
What is the colour of your skin (non exposed areas)?	Reddish	Very Pale	Pale with Beige tint	Light Brown	Dark Brown
Do you have freckles on unexposed areas?	Many	Several	Few	Incidental	Dark Brown
				Total Score for Genetic Disposition:	

Reaction to sun exposure					
Score	**0**	**1**	**2**	**3**	**4**
What happens when you stay in the sun too long?	Painful redness, blistering, peeling	Blistering followed by peeling	Burns sometimes followed by peeling	Rare burns	Never had burns
To what degree do you turn brown?	Hardly or not at all	Light colour tan	Reasonable tan	Tan very easy	Turn dark brown quickly
Do you turn brown within several hours after sun exposure?	Never	Seldom	Sometimes	Often	Always
How does your face react to the sun?	Very sensitive	Sensitive	Normal	Very resistant	Never had a problem
				Total Score for Genetic Exposure:	



African Health Sciences
Makerere Medical School

Afr Health Sci. 2023 Jun;23(2):753–763. doi: 10.4314/ahs.v23i2.86

Using artificial intelligence on dermatology conditions in Uganda: a case for diversity in training data sets for machine learning

Louis Kamulegeya [1], John Bwanika [1], Mark Okello [2], Davis Rusoke [1], Faith Nassiwa [2], William Lubega [1], Davis Musinguzi [1], Alexander Börve [3]

- First-Derm AI app for diagnosing dermatology conditions
- The app developer reported 70% accuracy for training results
- When tested in Uganda on 123 images, had 17% accuracy
- More accurate for diagnosing dermatitis than for fungal conditions

PASSION Project

Dataset

Sample Images



HEALTH TECH

STAT+

‘We need to be much more diverse’: More than half of data used in health care AI comes from the U.S. and China



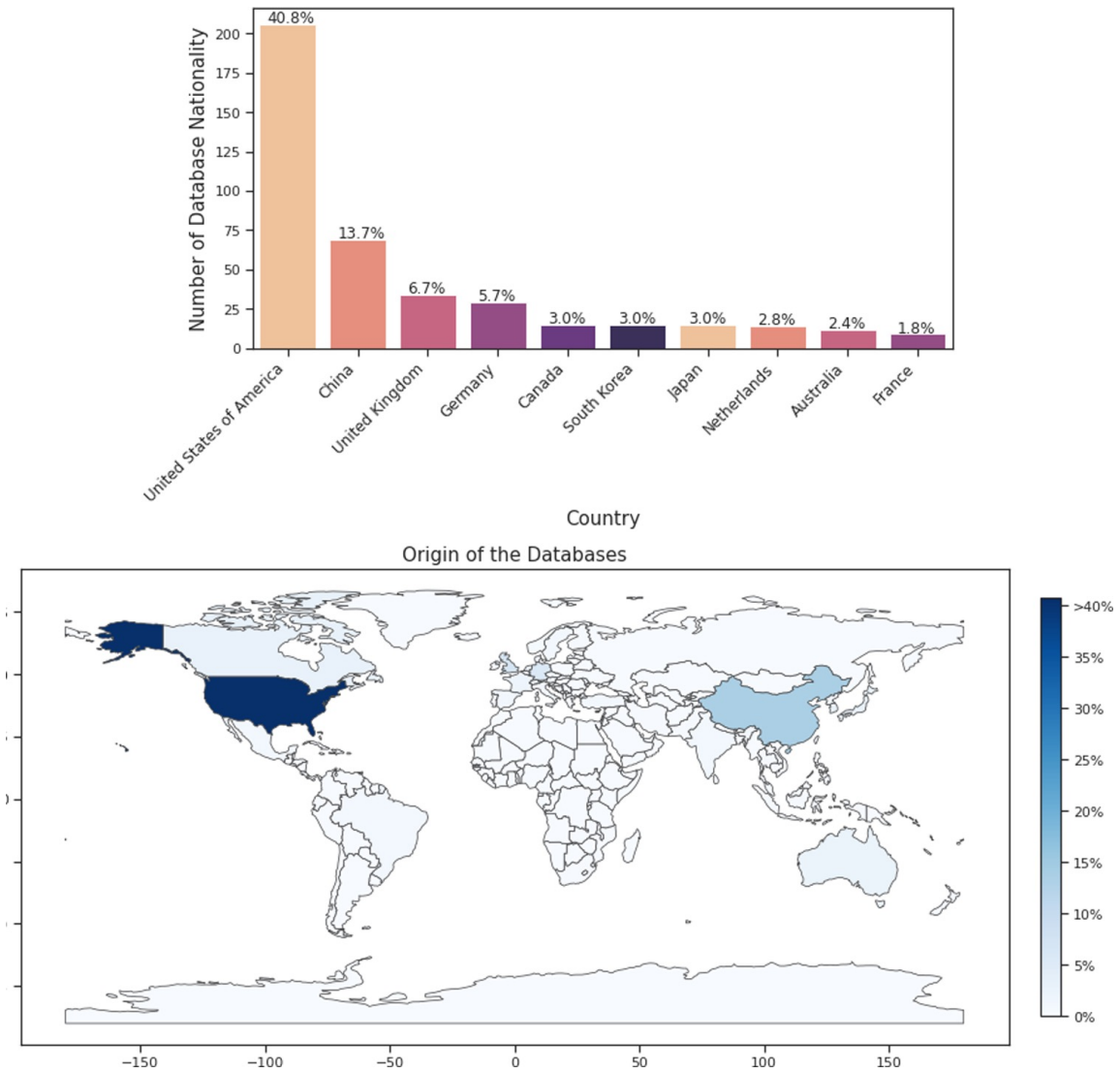
By Katie Palmer April 6, 2022

Reprints

PLOS Digit Health. 2022 Mar 31;1(3):e0000022. doi: 10.1371/journal.pdig.0000022

Sources of bias in artificial intelligence that perpetuate healthcare disparities—A global review

Leo Anthony Celi [1,2,3], Jacqueline Cellini [4], Marie-Laure Charpignon [5], Edward Christopher Dee [6], Franck Dernoncourt [7], Rene Eber [8], William Greig Mitchell [9,*], Lama Moukheiber [10], Julian Schirmer [8], Julia Situ [11], Joseph Paguio [12], Joel Park [13], Judy Gichoya Wawira [14], Seth Yao [12]; for MIT Critical Data



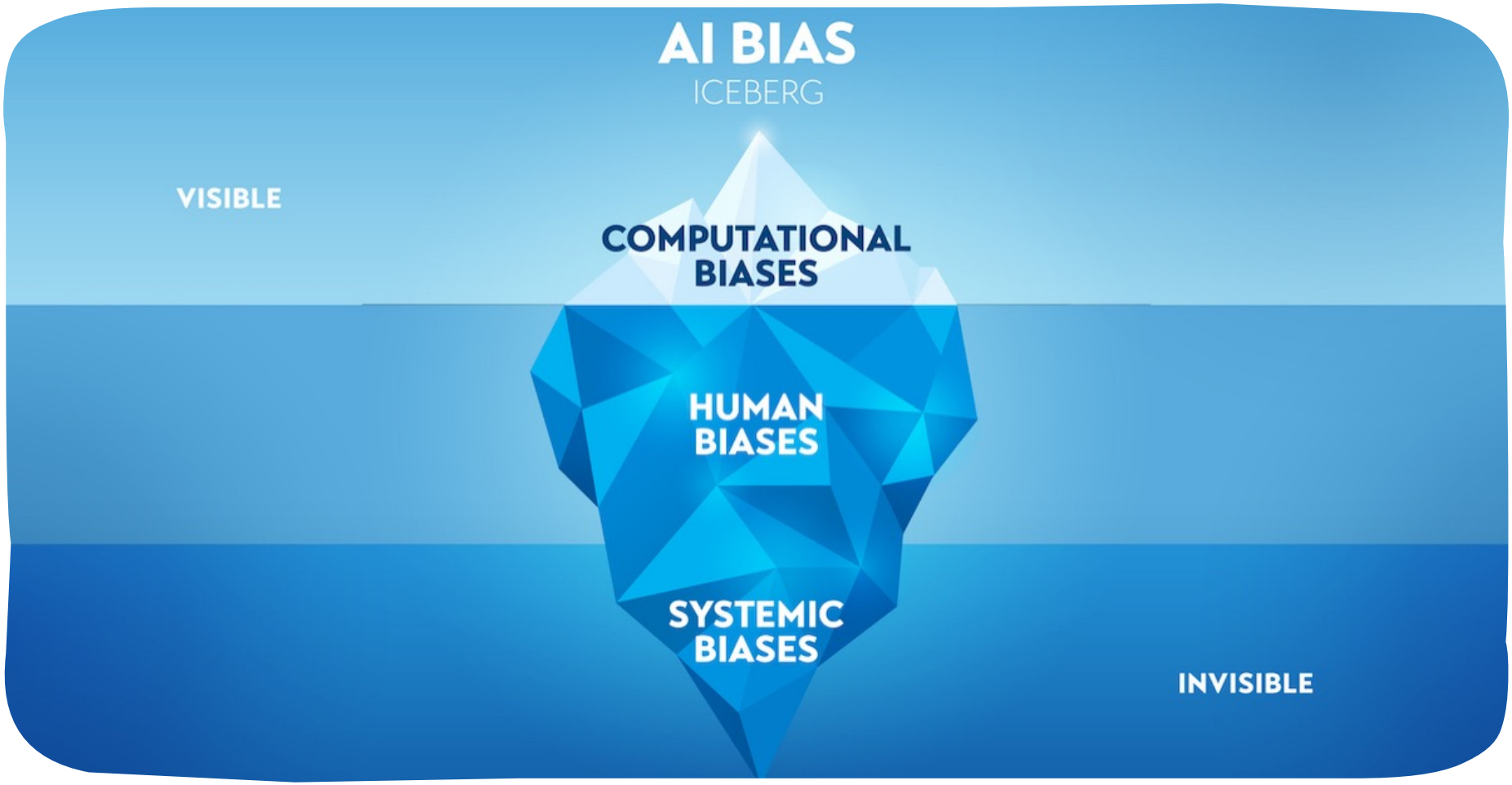
Study of 300 papers that used clinical AI and their associated datasets
(a) Distribution of overall database nationality in AI in medicine. (b) Heatmap of distribution of overall database nationality in AI in medicine (reference #35)

Randomised controlled trials evaluating artificial intelligence in clinical practice: a scoping review

Ryan Han, Julián N Acosta, Zahra Shakeri, John P A Ioannidis, Eric J Topol, Pranav Rajpurkar**

THE LANCET
Digital Health

- Of the 86 RCTs for AI assessed, 63% were assessed in a single center/location
- Need “more international collaboration and multicentre trials to ensure the generalizability of AI systems across various populations and health-care systems”



https://dhinsights.org/news/bias-in-ai

Strategies to Mitigate Bias

- Technical solutions (e.g. synthetic data; different weighting of data)
- Efforts to create more representative datasets
  - Issue of commercial datasets
- Evaluate within local context before implementation
- Equity in Design – practices throughout development process
- Engaging stakeholders

Process for Fairness




Co-designing AI with Kenya's health
needs in mind

How might different frameworks apply?

Ubuntu

- Communalism and openness
- Harmony and support
- Research prioritization
- Community empowerment and community-oriented decision-making

General AI ethical principles in US/Europe

- Privacy
- Transparency
- Equity/fairness
- Accountability/security

Bias – how to implement those principles?

- Does team have agreement on meaning of these terms?
  - Fairness means...
- Who on the team is implementing these principles?
  - Responsibility
    - Distribution of responsibility
- Evaluation

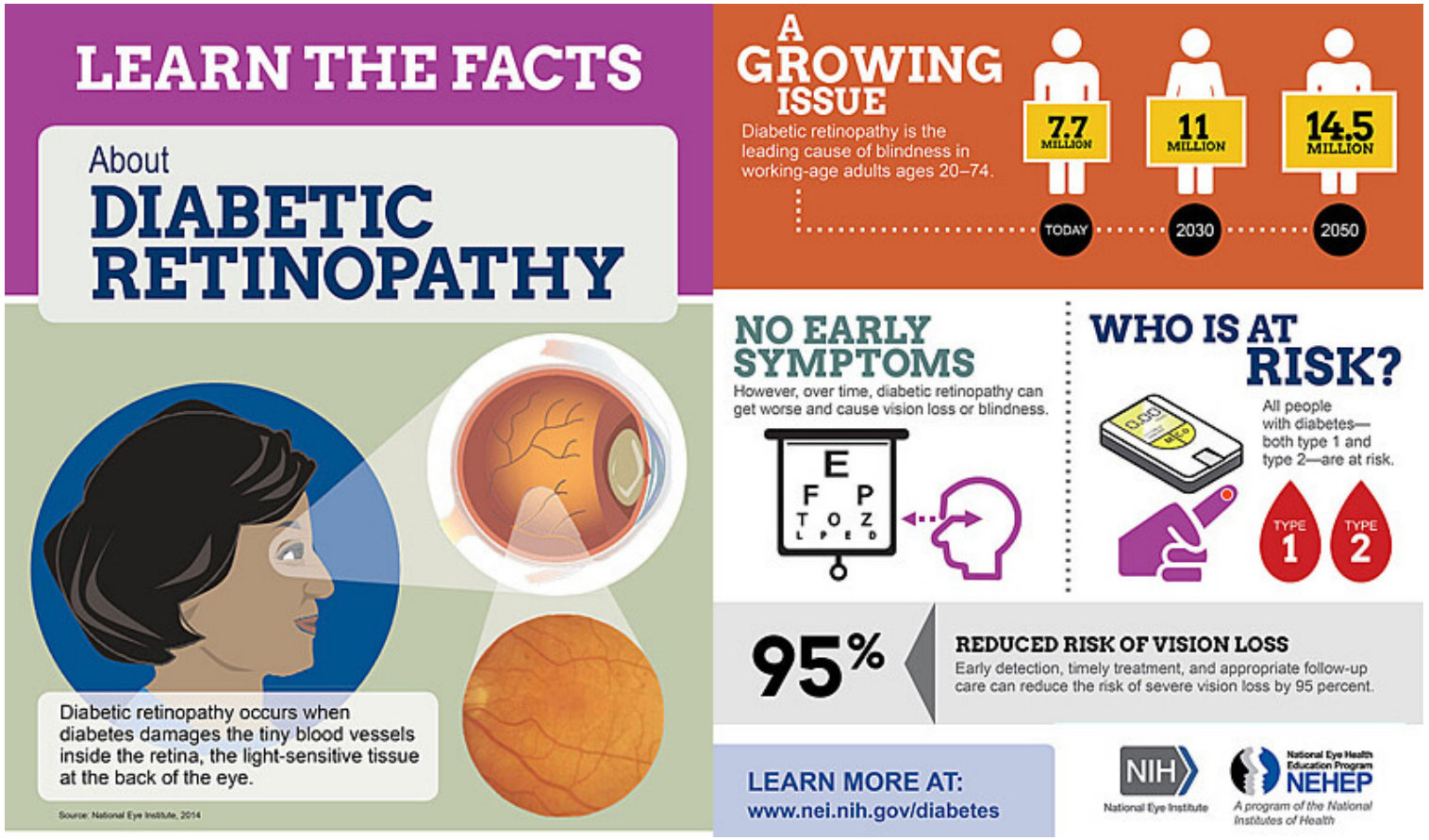
Defining Representativeness for Datasets

Figure 1: Varied meanings of fairness, bias, diversity and representation

	Fairness	Bias	Diversity	Representation
General definition	Similar treatment of individuals or groups in similar situations or with similar characteristics	Systematic error that impacts individuals or groups unfairly	Presence of data or individuals with a variety of identity or experiential backgrounds, with emphasis on historically underrepresented groups	Inclusion of diverse groups according to their proportions in a relevant population
Examples of different meanings or practices in biomedical research	Fairness as lack of algorithmic bias Fairness as taking steps to ensure fair outcomes across groups	Discrimination against underrepresented groups Unfair outcomes across groups Difference in algorithmic performance between groups	Inclusion of data according to a variety of genetic ancestry markers or self-reported races and ethnicities Workforce inclusion of individuals from underrepresented groups	Proportionally sampling data to match a regional population according to gender and disease status Hiring researchers from historically underrepresented backgrounds in all team roles

What are reasons for or against the statement that providing access to health AI is needed to avoid the kinds of disadvantages that are so systematically oppressive that people cannot have decent lives?

Diabetic retinopathy



Prevalence of Diabetic retinopathy

- In 2020, the number of adults worldwide with DR estimated to be 103.12 million
- Prevalence highest in Africa (35.90%) & North American and the Caribbean (33.30%) and was lowest in South and Central America (13.37%)



nature communications

Article

https://doi.org/10.1038/s41467-023-44676-z

Autonomous artificial intelligence increases screening and follow-up for diabetic retinopathy in youth: the ACCESS randomized control trial

Received: 16 June 2023

Accepted: 21 December 2023

Published online: 11 January 2024

Risa M. Wolf[1], Roomasa Channa[2], T. Y. Alvin Liu[3], Anum Zehra[1], Lee Bromberger[1], Dhruva Patel[1], Ajaykarthik Ananthakrishnan[1], Elizabeth A. Brown[1], Laura Prichett[4], Harold P. Lehmann[5] & Michael D. Abramoff[6,7,8,9,10]

Autonomous AI increased diabetic eye exam completion rates in youth with diabetes

- RCT randomized youth (ages 8-21 years) with type 1 and type 2 diabetes to intervention (autonomous AI diabetic eye exam at the point of care), or control (scripted eye care provider referral and education)
- Diabetic eye exam completion rate was significantly higher (100%,) in the intervention group (*n* = 81) than the control group (*n* = 83) (22%).
- In the intervention arm, 25/81 participants had an abnormal result, of whom 64% (16/25) completed follow-through with an eye care provider, compared to 22% in the control arm

› Lancet Digit Health. 2019 May;1(1):e35-e44. doi: 10.1016/S2589-7500(19)30004-4. Epub 2019 May 2.

Artificial intelligence using deep learning to screen for referable and vision-threatening diabetic retinopathy in Africa: a clinical validation study

Valentina Bellemo [1], Zhan W Lim [2], Gilbert Lim [2], Quang D Nguyen [1], Yuchen Xie [1], Michelle Y T Yip [3], Haslina Hamzah [1], Jinyi Ho [1], Xin Q Lee [1], Wynne Hsu [2], Mong L Lee [2], Lillian Musonda [4], Manju Chandran [5], Grace Chipalo-Mutati [6], Mulenga Muma [7], Gavin S W Tan [8], Sobha Sivaprasad [9], Geeta Menon [5], Tien Y Wong [8], Daniel S W Ting [10]

Concerns regarding AI for DR & Inequities

- Expert panel: “Because existing inequities will likely impact any potential benefit of the AI tool for specific populations, there needs to be initial investment in understanding those inequities before approving use of AI tools in different populations”

Expert panel on ethics of AI for DR:

- “Billions of dollar investments into AI mean billions of dollars not invested in universal health coverage.”
- “It shows how technology is always only one piece of a larger problem. The ethical issue then is insufficient scientific practices contribute to research waste – e.g., if focus only on the model's performance rather than a valued outcome which speaks to social value.”



Expert panel on AI for DR: Access

"...access is a major problem for implementation - will the tool, if built upon research in this population, make its way back for use by this population? if the population does not have resourcing to pay for the screening and/or treatment, then are they reaping the benefits of participating in the R&D?"

Need for addressing equity in development as well as for evaluating outcomes

frontiers in Digital Health

Front Digit Health. 2020 Jul 7;2:6. doi: 10.3389/fdgth.2020.00006

Artificial Intelligence for Healthcare in Africa

Ayomide Owoyemi [1,*], Joshua Owoyemi [2], Adenekan Osiyemi [3], Andy Boyd [1]

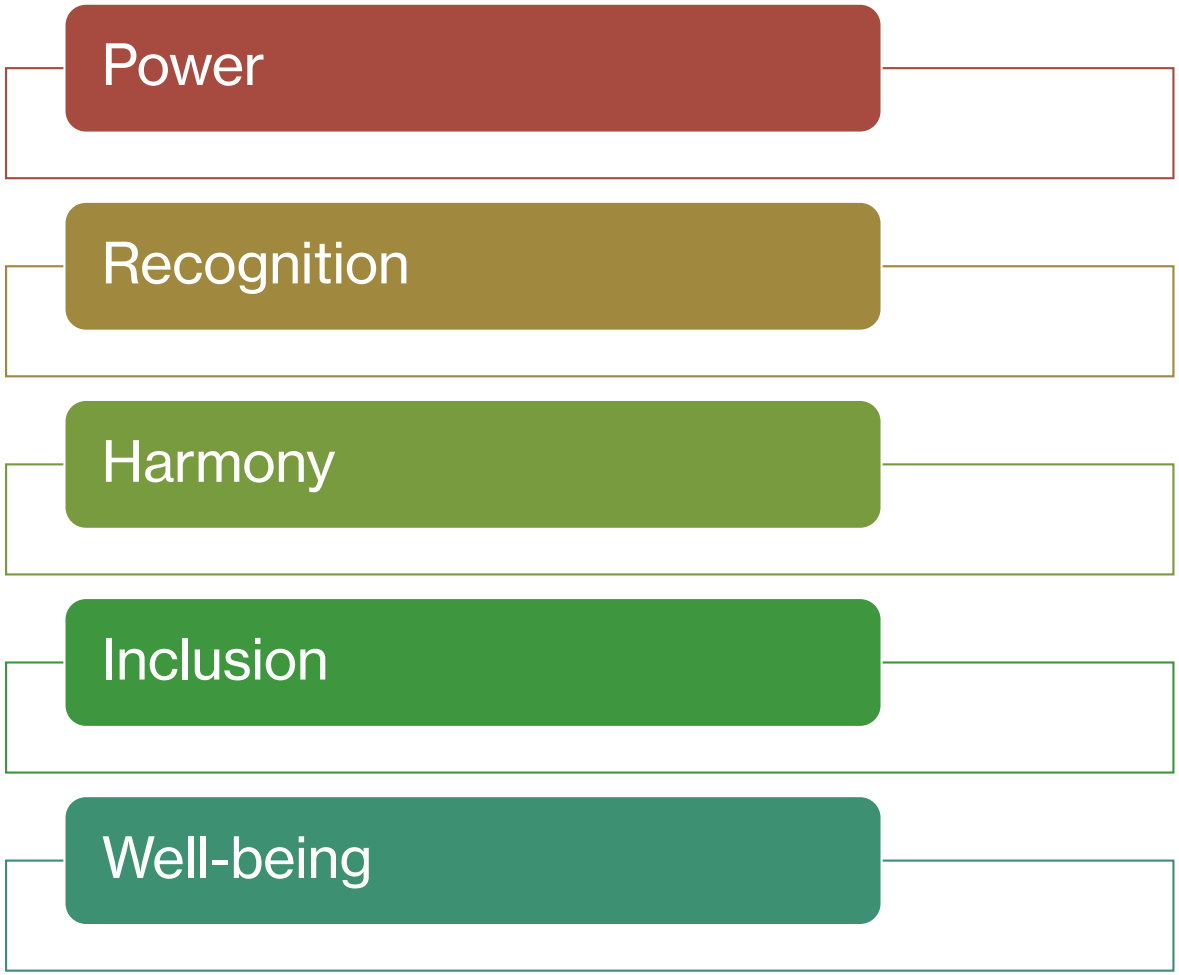
- Challenges include:
  - Costs of data acquisition & processing
  - Hardware and computing tech costs (e.g. cloud computing)
  - System maintenance
  - Infrastructure (e.g. internet, electricity)

Next week: Data Justice

Goals

- Understand what surveillance is, its risks and potential benefits, and how it is practiced in different global contexts
- Analyze the social and ethical impacts of data access and inequities across global contexts

Five core dimensions of social justice (Pratt)



Adoption of AI in Ghanaian Hospitals

